where $R_0 = \lambda N_0$

The equation (9.37) is also equivalent to radioactive law of decay. Here R_0 is the activity of the sample at *t=0* and *R* is the activity of the sample at any time *t*. From equation (9.37), activity also shows exponential decay behavior. The activity *R* also can be expressed in terms of number of undecayed atoms present at any time *t*.

From equation (9.37), since $N = N_0 e^{-\lambda t}$, we write

$$R = \lambda N \qquad (9.38)$$

Equation (9.38) implies that the activity at any time *t* is equal to the product of decay constant and number of undecayed nuclei present at that time *t*. Since *N* decreases with time, *R* also decreases.

The SI unit of activity *R* is Becquerel and one Becquerel (Bq) is equal to one decay per second. There is also another standard unit for the activity called Curie(Ci).

1 Curie =1 Ci = 3.7×10^{10} decays per second

1 Ci = 3.7×10^{10} Bq

Note

Initially one curie was defined as number of decays per second in 1 g of radium and it is equal to 3.7×10^{10} decays/s.

9.6.5 Half-life

It is difficult to calculate the time taken by a given a sample of N atoms to decay completely. However, we can calculate the time taken by the given sample of atoms to reduce some fraction of the initial amount.

We can define the half-life $T_{1/2}$ as the time required for the number of atoms initially present to reduce to one half of the initial amount.

The half-life is the important characteristic of every radioactive sample. Some radioactive nuclei are known to have half-life as long as 10^{14} years and some nuclei have very shorter half-life time (10^{-14}s).

We can express half-life in terms of the decay constant. At $t = T_{1/2}$, the number of undecayed nuclei $N = \frac{N_0}{2}$.

By substituting this value in to the equation (9.35), we get

$$\frac{N_0}{2} = N_0 e^{-\lambda T_{1/2}}$$

$$\frac{1}{2} = e^{-\lambda T_{1/2}} \text{ or } e^{\lambda T_{1/2}} = 2$$

Taking logarithm on both sides and rearranging the terms,

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.6931}{\lambda} \qquad (9.39)$$

Note

One should not think that shorter half-life material is safer than longer half-life material because it will not last long. The shorter half-life sample will have higher activity and it is more 'radioactive' which is more harmful.

If the number of atoms present at t=0 is N_0, then $\frac{N_0}{2}$ atoms remain undecayed in first half-life and $\frac{N_0}{4}$ atoms remain undecayed after second half life and so on. In general, after *n* half-lives, the number of nuclei remaining undecayed is given by

$$N = \left(\frac{1}{2}\right)^n N_0 \qquad (9.40)$$

where *n* can be integer or non-integer. Since the activity of radioactive sample also

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